

Luca Drole

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Biomedical engineer developing robust and data-efficient learning methods for medical imaging. My work focuses on representation learning, Vision Transformers, and generative models to address domain shift and data scarcity in MRI. Committed to principled, reproducible research enabling clinically reliable AI systems.

EDUCATION

ETH Zürich <i>M.Sc. Biomedical Engineering (Biomedical Imaging) – 5.5/6</i> - Core areas: Imaging modalities, Machine Learning, Medical Image Analysis - Top 60 incoming Master's students in 2023	Switzerland 2023 – Present
Politecnico di Milano <i>B.Sc. Biomedical Engineering – 110/110 cum Laude</i> Course Representative; Student Researcher at NECSTLab	Italy 2020 – 2023
Liceo Scientifico L.S. Oberdan, Trieste <i>Scientific Lyceum – 100/100 cum Laude</i>	Italy 2015 – 2020

RESEARCH EXPERIENCE

Master's Thesis – Vision Transformers for Medical Imaging <i>ETH Zürich (Computer Vision Laboratory)</i> - Designing data-efficient Vision Transformer architectures for medical image segmentation - Developing scalable GPU training pipelines on Slurm-based HPC with experiment tracking (W&B) - Supervisor: Dr. Ertunc Erdil	Oct 2025 – Present
Visiting Research Student <i>Martinos Center for Biomedical Imaging, MGH & Harvard Medical School</i> - Developed contrast-agnostic MRI feature extraction backbone using 3D ADNI data (135 subjects) - Reduced brain-age prediction error by 20% across heterogeneous MR datasets - Supervisors: Dr. Peirong Liu, Dr. Xiaoling Hu	Mar 2025 – Sept 2025 Boston, MA
Semester Project – Cardiac MRI Scar Segmentation <i>ETH Zürich (Cardiac Magnetic Resonance Group)</i> - Built CycleGAN-based framework for LGE-MRI to scar-mask translation (38 patients) - Produced scar segmentations without explicit supervision, with a median Dice score of 0.88 - Benchmarked GAN architectures in PyTorch and TensorFlow - Supervisors: I. Margolis, Dr. L. Dal Toso	Feb 2024 – Jun 2024
Bachelor's Thesis – Histopathology Segmentation <i>NECSTLab, Politecnico di Milano</i> - Developed class-imbalance-aware loss for Hover-Net nuclei segmentation - Implemented complete PyTorch training pipeline - Supervisors: I. Poles, Dr. E. D'Arnese	Feb 2023 – Jul 2023
Research Intern (Amgen Scholar) <i>ETH Zürich (Laboratory for Movement Biomechanics)</i> - Automated joint implant pose estimation from fluoroscopy videos using classical CV + DL - Developed autoencoder network to impute implant poses from occluded frames - Runner-up Best Poster Prize, Amgen Scholars European Symposium (University of Cambridge) - Supervisor: Dr. Florian Vogl	Jul 2022 – Sept 2022
Research Intern – EEG-fMRI Analysis <i>Polyclinic Hospital Milan & Politecnico di Milano</i> - Multimodal EEG-fMRI analysis of patients with Major Depressive Disorder - Supervisor: Prof. E. Maggioni	Nov 2022 – Jun 2023

CONFERENCE PAPER

- **Drole, L.**, Poles, I., D'Arnese, E., Santambrogio, M. D. *Towards a Lightweight 2D U-Net for Accurate Semantic Segmentation of Kidney Tumors in Abdominal CT Images*. In: IEEE EUROCON 2023 – International Conference on Smart Technologies, 2023.

TEACHING

Teaching Assistant

Sept 2024 – Jan 2026

ETH Zürich (D-ITET, D-BAUG)

- TA for “Programming for Engineers” and “Biomedical Imaging” (grading, tutorials, course material updates)

LEADERSHIP

Junior Board Member for Fundraising

Oct 2020 – Nov 2022

European Youth Parliament Italy

- Led partnerships and sponsorship acquisition; coordinated national grant applications
- Contributed to organisational governance and long-term strategy

Project Manager

Jan 2021 – Nov 2021

European Youth Parliament Italy, Trieste

- Organised a four-day international European Parliament simulation with 100+ participants
- Managed budget, fundraising, logistics, academic coordination, and 40+ volunteers

AWARDS

- ETH Zürich Excellence Scholarship (CHF 48,000) – Global **Top 60** incoming Master’s students
- Nova 111 Italy (2025) – Top 10 Engineering & Technology Students Nationwide
- Runner-up Best Poster Prize – Amgen Scholars European Symposium
- Best Freshmen Award – Politecnico di Milano
- National Excellence Honours Roll – Italian Ministry of Education

VOLUNTEERING & ORGANISATIONS

- European Youth Parliament (2018 – Present): Academic facilitator, team leader, and trainer across 30+ international educational events; directed policy debates on EU Health, Technology, and Energy legislation; currently serving on the Board of Guarantors

TECHNICAL SKILLS

Languages: Python, MATLAB

ML Frameworks: PyTorch, TensorFlow

Deep Learning: Vision Transformers, CNNs, GANs, Diffusion Models

Medical Imaging: MRI, CT, Histopathology, Multimodal Analysis

Tools: Git, Docker, Slurm, Linux, LaTeX

LANGUAGES

- Italian (Native), English (C2 – IELTS 8.5), German (A2)